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**The relationship between trading in the
S&P 500 index and the frequency of
searches for the words "stocks" and
"Trump"**

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1 Abstract

In this seminar paper, we will study the links between trading in the S&P 500 index and searches for the terms "Trump" and "stocks" on Google. The S&P 500 index consists of the 500 most valuable stocks actively traded in the United States, and American politics often has a major influence on its value.

We will investigate the relationship of hml (the difference between the highest and lowest prices during the day, *high minus low*), change (the closing price minus the opening price during the day), and volume (the quantity of the index traded during the day) with the relative frequencies of searches for these terms on Google. Using descriptive statistics, we will examine the distributions of these quantities and formulate hypotheses based on them.

We will then use linear regression based on the frequency of searches for the keywords to estimate certain indicators of the S&P 500 index. We will also test the hypothesis that the distribution of HML is log-normal. This analysis will establish the nature of the relationship between trading in the S&P 500 index and the use of the search engine.

2 Introduction

2.1 Data on the S&P 500 index

To obtain reliable historical data on movements in the price of the S&P500 index, we decided to use Yahoo Finance. Specifically, we examined the *ticker* ^GSPC, which tracks only the price of the index (excluding dividend payments). Although indices are not financial instruments (they cannot be traded), Yahoo Finance provides its own internal estimate of the trading volume of financial instruments that track the S&P500 (ETFs, futures, and similar instruments).

For the ticker ^GSPC, we examined historical data from 3 October 2024 to 30 April 2025. We chose October as the starting point to include the period before the election, during which there were many interesting market movements.

Specifically, the data we downloaded for these days were:

- price at market opening
- highest price during the day
- lowest price during the day
- price at market closing
- total volume for the day

2.2 Search data

For search data, we decided to use Google Trends. Google is unquestionably the most widely used search engine today, so its data are the most relevant. Although Google Trends does not provide access to absolute frequencies (the exact number of searches), it does provide relative data. More precisely, the data are normalized so that the highest frequency is exactly 100. In addition, all values are integers. Google Trends allows search results to be examined within a particular region. Since the S&P500 is an American market index, we also decided to examine search data only for the United States.

We examined the terms "stocks" and "Trump" and, to obtain the most detailed data possible, retrieved the data separately (so that each term would be normalized separately). We consider the normalization used by Google Trends unproblematic for examining linear dependence, since it is itself a linear transformation.

3 Descriptive statistics

3.1 Data on the S&P 500 index

From the data obtained, we will derive three different quantities to examine.

3.1.1 Maximum change

We will call the first quantity maximum change or hml (*high-minus-low*), and calculate it using the formula:

$$\text{hml} = (\text{highest price during the day}) - (\text{lowest price during the day})$$

Its representation as a time series is:

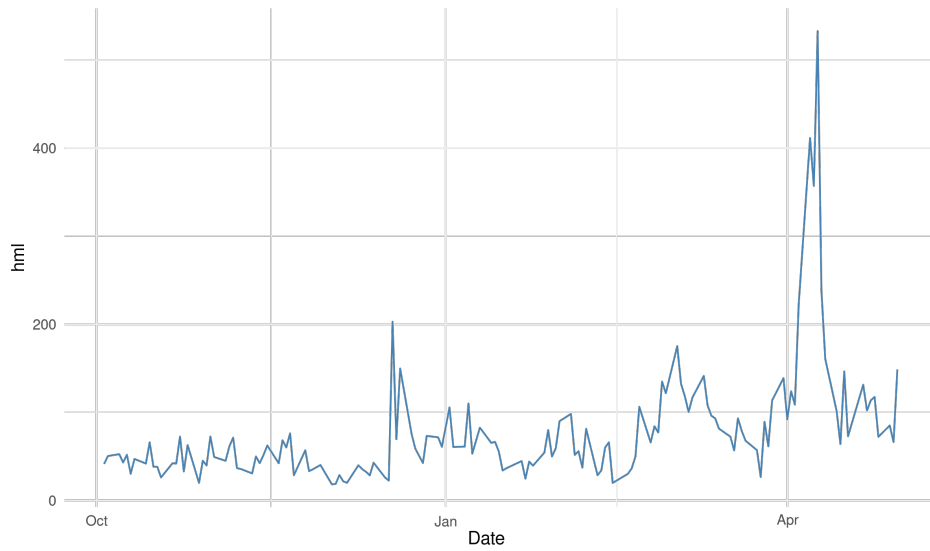


Figure 3.1: hml during the period examined

The five-number summary is:

$$(x_0(v), q_L(v), m(v), q_U(v), x_{(n)}(v)) = (18.25, 40.80, 60.88, 91.01, 532.91)$$

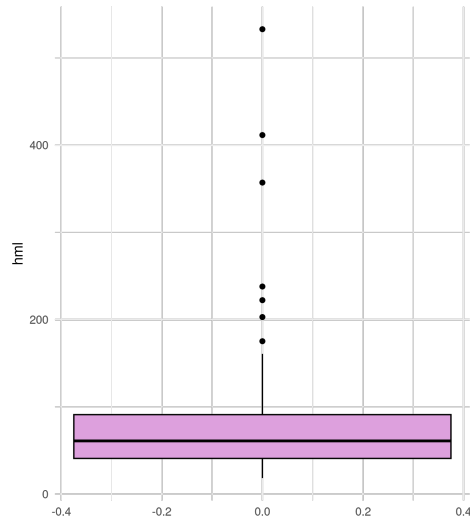
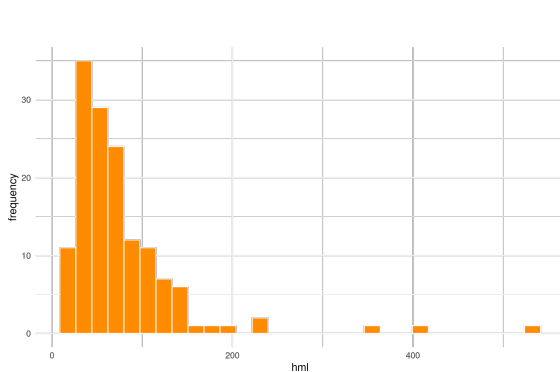


Figure 3.2: Boxplot

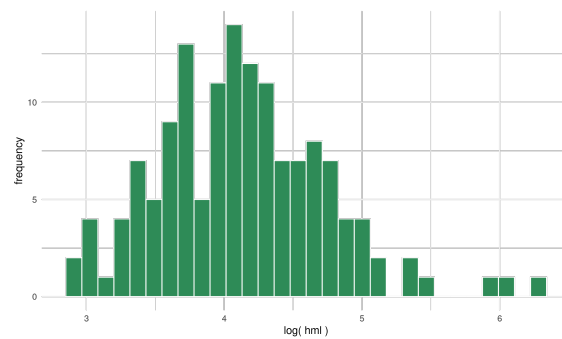
The boxplot shows many outliers, which is to be expected. During the period examined, several events had a huge effect on the market, including:

- Trump's election as president
- Trump's inauguration
- Trump's introduction of tariffs

A natural question is which distribution the hml of a trading day comes from. To investigate this further, we plotted its histogram and the histogram of its logarithm. This produced the following plots:



(a) Histogram of hml during the period examined



(b) Histogram of $\log(\text{hml})$ during the period examined

Figure 3.3: Histograms of hml

After examining panel (b), we decided to test whether hml follows a log-normal distribution. Specifically, we performed a test for the MLE given by the parameters

$$\hat{\mu} = 4.1276 \quad \hat{\sigma}^2 = 0.3814$$

The distribution is shown below, overlaid with the theoretical curve:

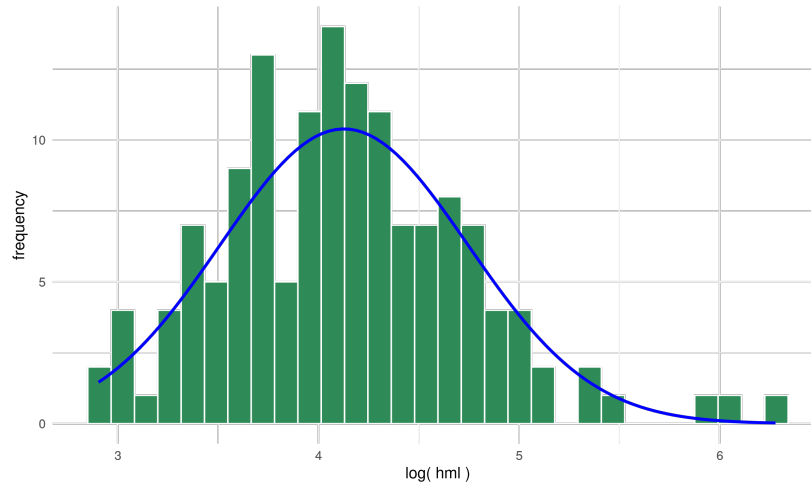


Figure 3.4: Histogram of logarithmic values with the appropriately scaled curve

3.1.2 Volume

The next quantity we examined was volume. We did not calculate volume because it was provided directly in the table downloaded from Yahoo Finance. As mentioned above, this figure represents the total trading turnover of the S&P500 index and attempts to account for all ETFs tracking its value. As a result, the figures are enormous, even for the financial sector.

The representation of volume as a time series is:

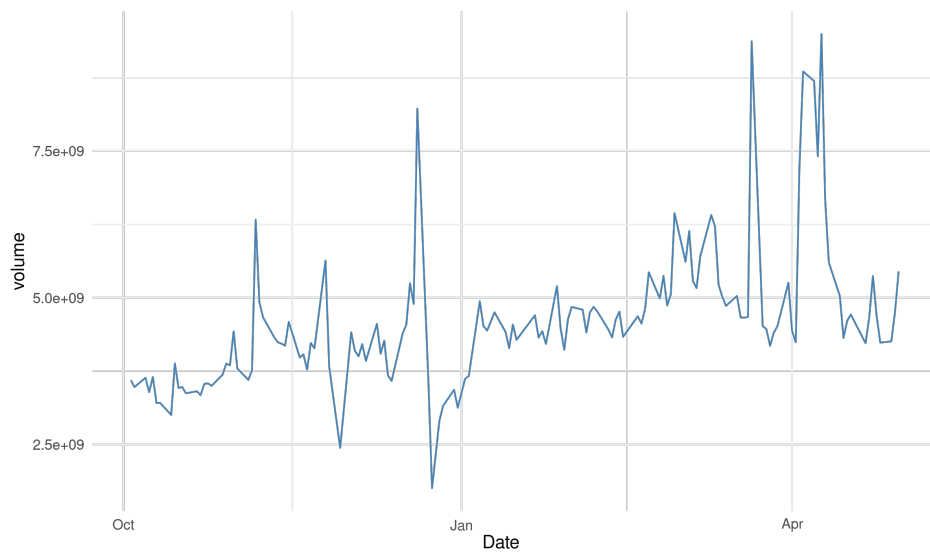


Figure 3.5: Volume during the period examined

The five-number summary is:

$$(x_0(v), q_L(v), m(v), q_U(v), x_n(v)) = (1757720000, 3880610000, 4432250000, 4866380000, 9489600000)$$

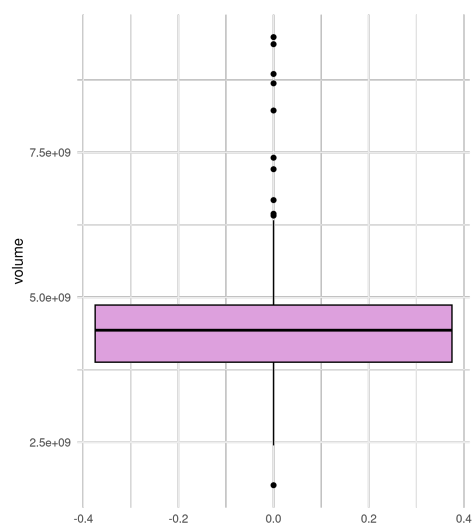


Figure 3.6: Boxplot

The boxplot again shows a large number of outliers, for reasons similar to those in the

previous example. To obtain a clearer picture of the empirical distribution, we plotted histograms:

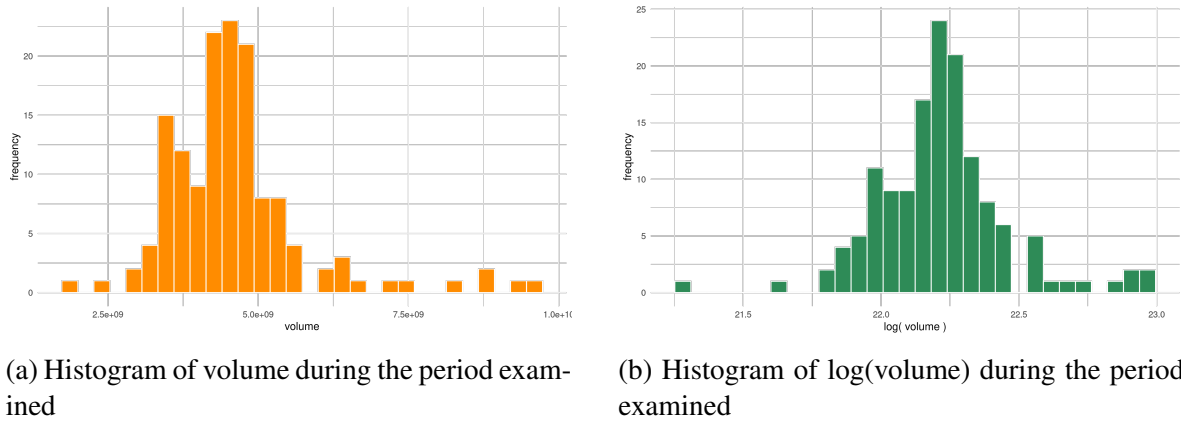


Figure 3.7: Histograms of volume

3.1.3 Change

The final quantity we examined is the change between the price at market opening and the price at market closing. We calculated it using the formula

$$\text{change} = (\text{closing price}) - (\text{opening price})$$

The representation of change as a time series is:

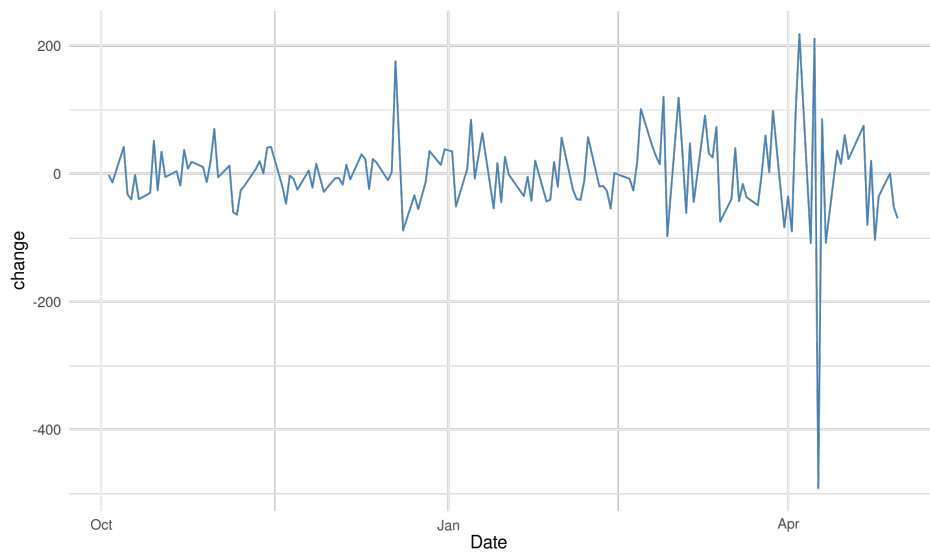


Figure 3.8: Change during the period examined

The five-number summary is:

$$(x_0(v), q_L(v), m(v), q_U(v), x_n(v)) = (-491.62012, -34.46997, -1.75000, 28.81982, 218.06006)$$

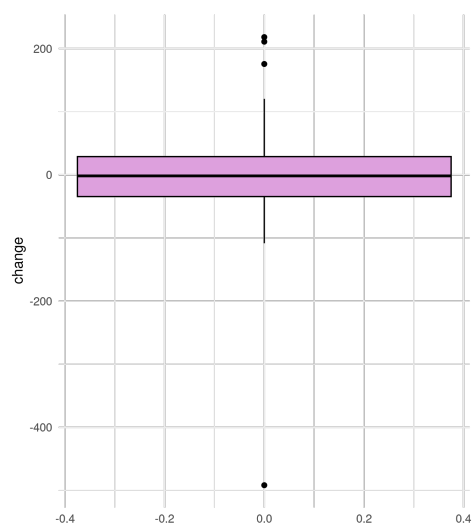


Figure 3.9: Boxplot

Because of the large outlier corresponding exactly to the day tariffs were introduced, we

decided to plot not only the ordinary histogram but also a histogram of the data from the 5th to the 95th percentile, to see the distribution of the central data more clearly.

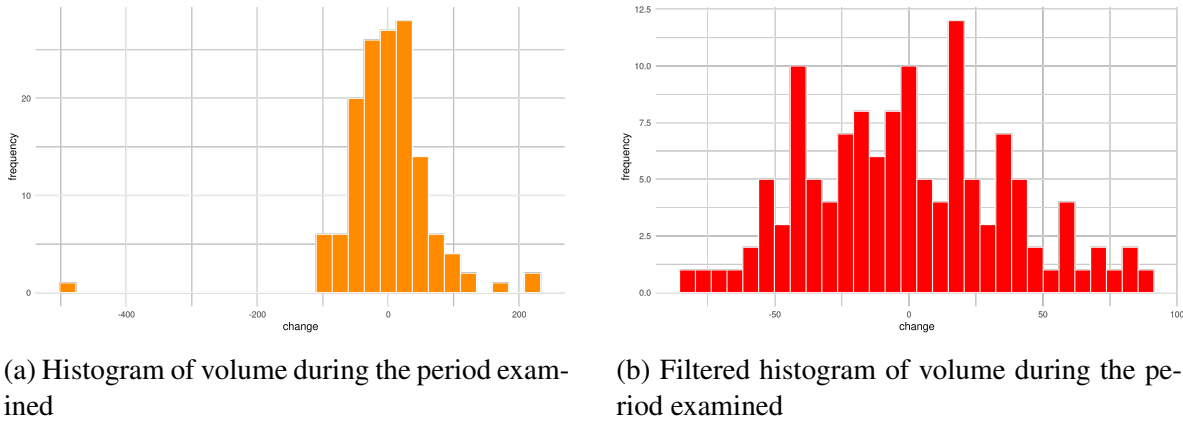


Figure 3.10: Histograms of volume

Based on the histogram, we decided to test whether change is normally distributed. Specifically, we performed a test for the MLE given by the parameters

$$\hat{\mu} = -0.9113 \quad \hat{\sigma}^2 = 4675.9851$$

The distribution is shown below, overlaid with the theoretical curve:

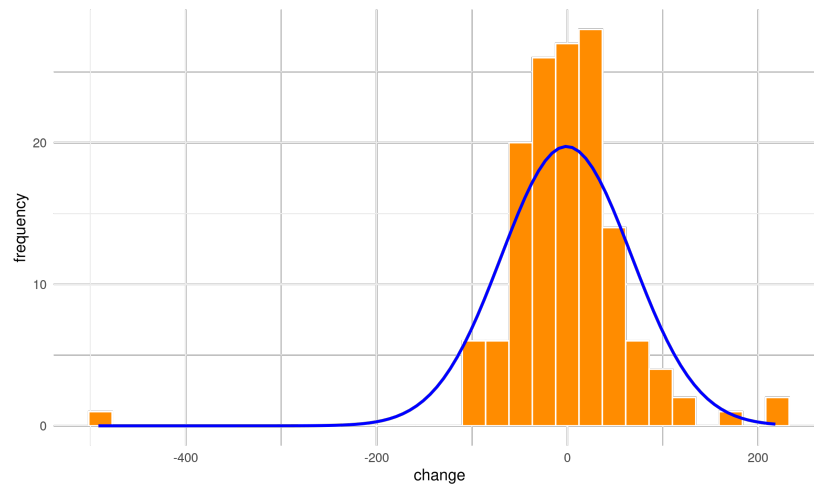
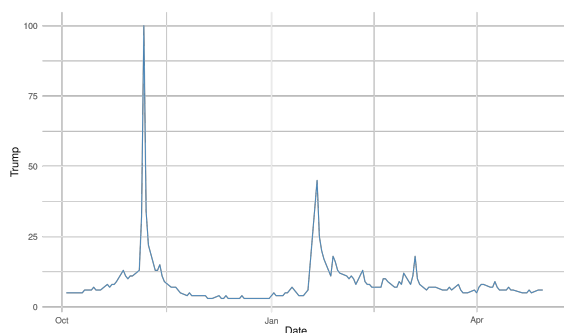


Figure 3.11: Histogram with the appropriately scaled curve

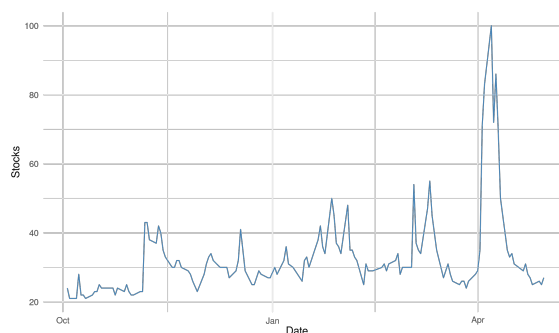
3.2 Search data

The data shown throughout represent **relative** frequencies: each frequency is divided by the highest frequency in the given time interval and multiplied by 100. The maximum value in both time series will therefore be exactly 100.

The data represented as time series are:



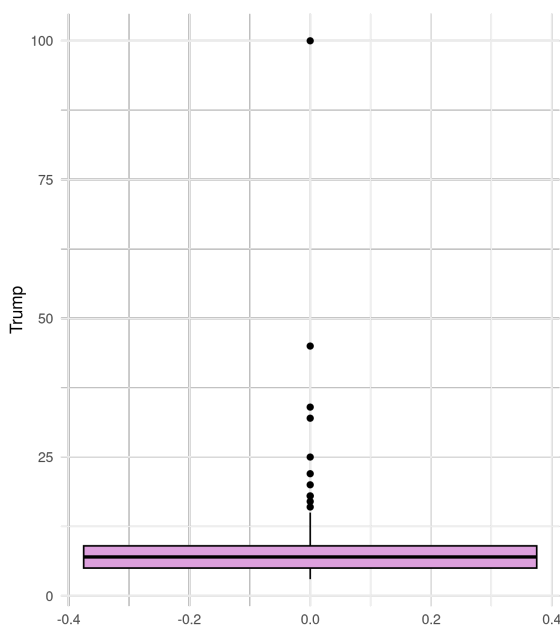
(a) The term "Trump"



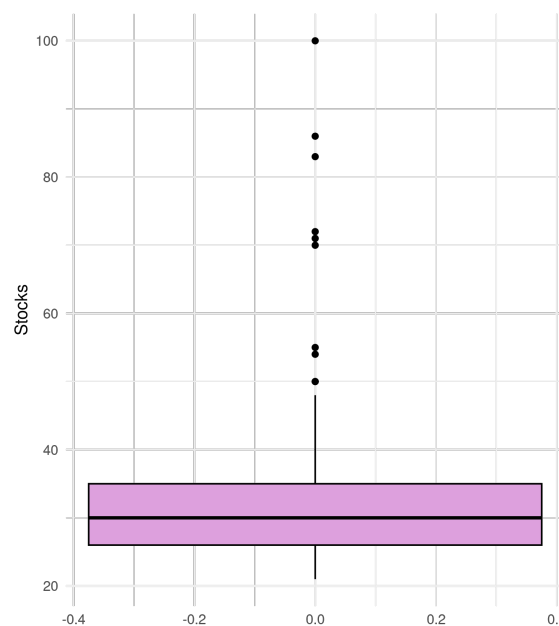
(b) The term "Stocks"

Figure 3.12: Time series

Their boxplots are



(a) The term "Trump"



(b) The term "Stocks"

Figure 3.13: Boxplots

The boxplots show that the outliers for the term "Trump" are more pronounced than those for "Stocks". The main reason is the US presidential election, held in November 2024. This corresponds exactly to the maximum frequency.

For the term "Stocks", we observe the maximum on the American "Liberation Day", the day President Trump announced his plan to introduce tariffs. Immediately afterwards, the S&P500 experienced the third-largest three-day decline in American history. Observing this phenomenon provided our initial motivation for this seminar paper.

The five-number summary for the term "Trump" is:

$$(x_0(v), q_L(v), m(v), q_U(v), x_n(v)) = (3, 5, 7, 9, 100)$$

For the term "Stocks", it is:

$$(x_0(v), q_L(v), m(v), q_U(v), x_n(v)) = (21, 26, 30, 35, 100)$$

3.3 Correlation coefficients

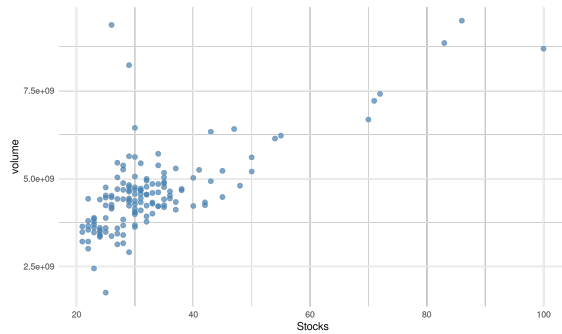
Finally, we decided to examine the correlation coefficients to determine which pairs of quantities to investigate in more detail. The resulting table of coefficients is shown below.



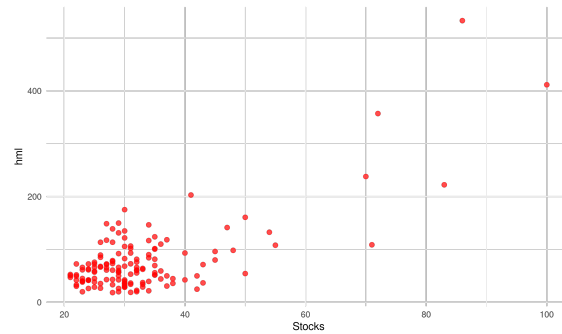
Figure 3.14: Correlation coefficient matrix

From this table, we concluded that change was not worth investigating much further and

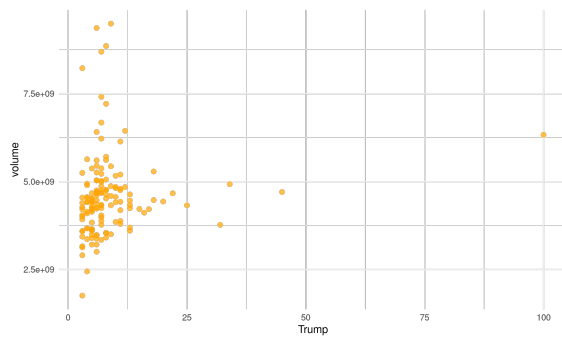
that the correlations with volume and hml were much stronger. Finally, we plotted four scatterplots for these pairs.



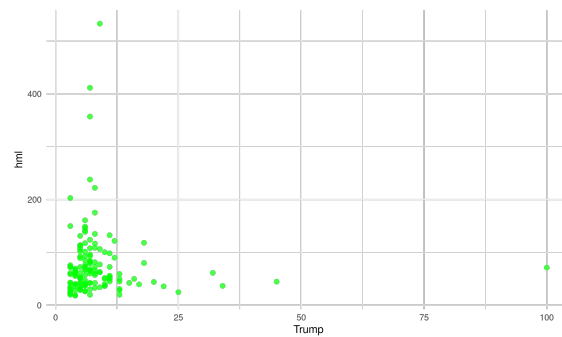
(a) Stocks — volume



(b) Stocks — hml



(c) Trump — volume



(d) Trump — hml

Figure 3.15: Scatterplots

4 Inferential statistics

4.1 Single-parameter linear regression

Let X denote the relative frequency of searches for the word "stocks", and let Y denote the trading volume of the S&P 500 index. Performing linear regression between these two variables, with X as the independent variable and Y as the dependent variable, gives the line $Y = \hat{\beta}X + \hat{\alpha}$, where $\hat{\alpha} = 2.274 \times 10^9$ and $\hat{\beta} = 7.038 \times 10^7$. Let us test the significance of this model at the 0.05 significance level. We take the hypotheses

$$H_0 : \beta = 0,$$

$$H_1 : \beta \neq 0,$$

but first check the Gauss-Markov conditions to show that the least-squares estimators are the best linear unbiased estimators.

To check the Gauss-Markov conditions, we will use the plots in Figures 4.1, 4.2, 4.3, 4.4. The first plot shows that the expected value of the residuals is approximately 0, regardless of the value predicted by linear regression. We therefore conclude that the expected values of the residuals are 0, meaning that the first Gauss-Markov condition is satisfied.

Figure 4.2 contains a Q-Q plot comparing the quantiles of the standardized residuals with those of the normal distribution. If the standardized residuals had a standard normal distribution, the points would follow the line $y = x$. Since the points lie approximately on a line through the origin, we can accept the assumption that the standardized residuals are normally distributed. A linear transformation of the residuals, specifically scaling, would produce a Q-Q plot with values lying on the line $y = x$. Since scaling a standard normal distribution produces a normal distribution centered at 0, the assumption is justified.

The third plot shows that the square root of the absolute value of the standardized residuals is approximately constant, at least in the left half, where most of the data lie. We conclude that the variances of the residuals are approximately constant and consider the second Gauss-Markov condition satisfied.

It remains to discuss the lack of correlation between the residuals. Since the data have a time component, we naturally expect observations to be correlated with those close to them in time. Reducing the granularity of the data would reduce the correlation because the observations would be further apart in time, but in this paper we will use daily granularity. We also justify the erratic nature of daily trading volume using the time series in Figure 3.5.

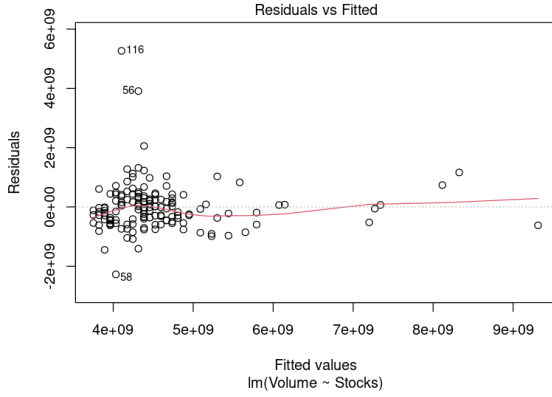


Figure 4.1: Relationship between $\hat{Y}$ and the residuals

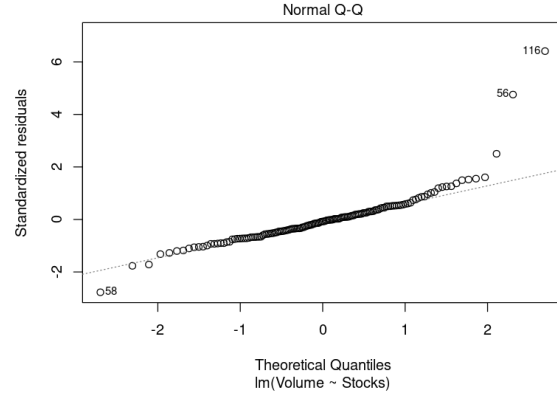


Figure 4.2: Q-Q plot of residuals

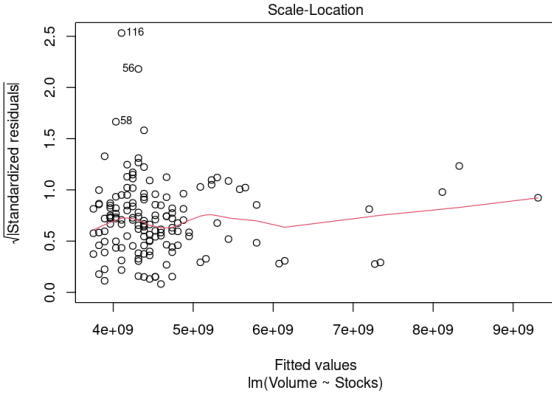


Figure 4.3: Relationship between $\hat{Y}$ and the standardized residuals

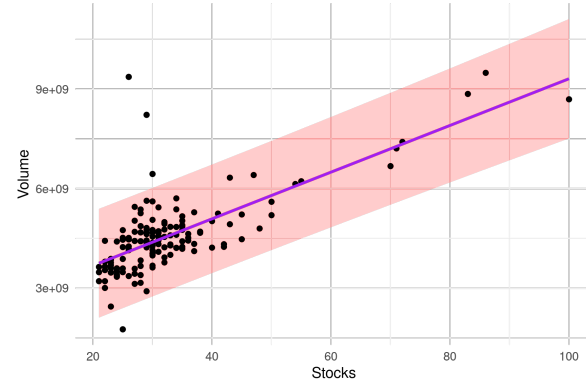


Figure 4.4: Linear regression line with a confidence interval

We can now construct a 95% confidence interval for β using the statistic

$$\frac{\hat{\beta} - \beta}{\hat{\sigma} \cdot \sqrt{\frac{1}{S_{xx}}}} \sim t(n - 2).$$

The resulting interval is

$[5.9165060 \times 10^7, 8.1599495 \times 10^7]$. This also leads us to conclude that the linear regression is statistically significant at the 0.05 level, since 0 is not within this interval; we reject the hypothesis H_0 . For α , the 95% confidence interval is $[1.880536685 \times 10^9, 2.666848278 \times 10^9]$.

Figure 4.4 illustrates the linear model with confidence intervals. We justify the relatively

close fit of the data to the linear model by the substantial R^2 value of 0.5218 and the high Pearson correlation coefficient in Table 3.14. It is interesting to note that the most notable outlier, observation 116, occurred on 21.03.2025., the day Trump announced that he would present a new tariff strategy on "Liberation Day".

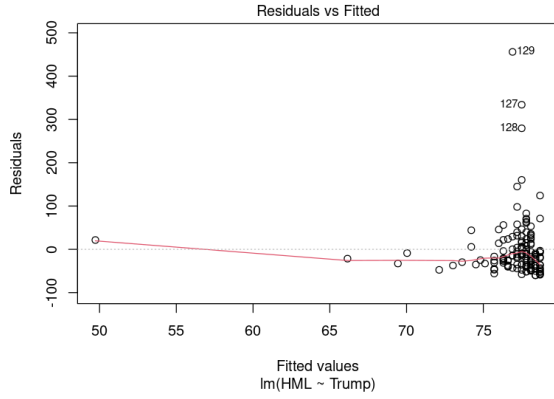


Figure 4.5: Relationship between $\hat{Y}$ and the residuals

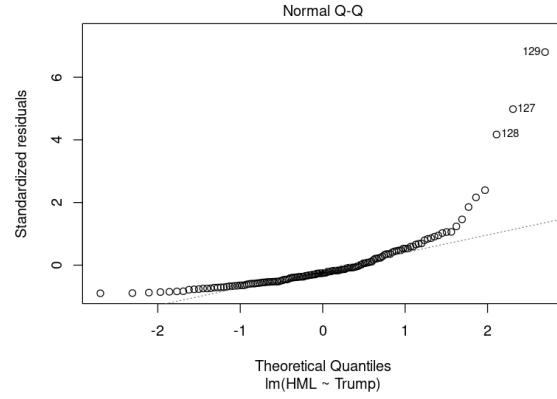


Figure 4.6: Q-Q plot of residuals

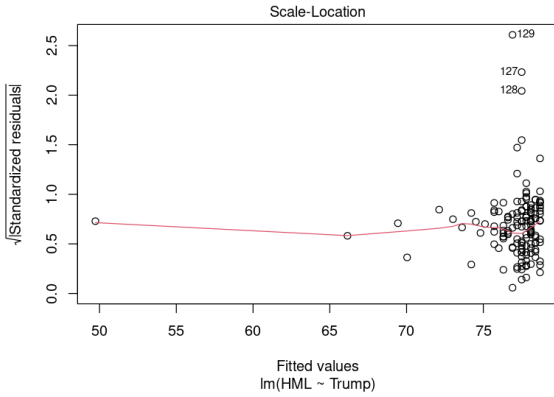


Figure 4.7: Relationship between $\hat{Y}$ and the standardized residuals

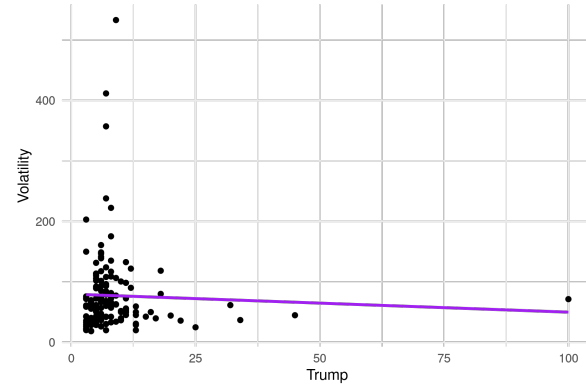


Figure 4.8: Linear regression line

Let us now consider the dependence of HML on "Trump". From the plots in Figures 4.5, 4.6, 4.7, 4.8, we see that the Gauss-Markov conditions are not satisfied. The expected value of the residuals decreases substantially at high values predicted by the linear model, and the Q-Q plot does not resemble a line through the origin at all. Nevertheless, Figure 4.8 suggests that it makes more sense to perform linear regression between the logarithms of the two variables, since both variables have "long tails" that can be tamed using logarithms.

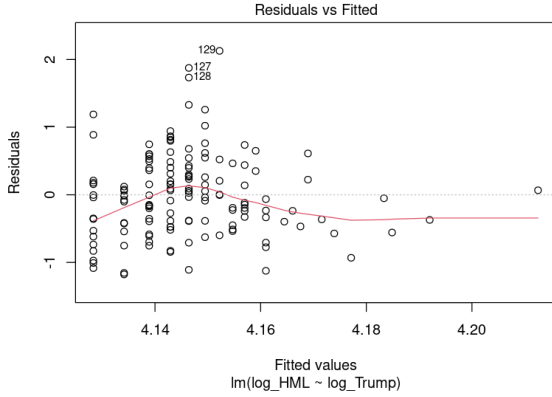


Figure 4.9: Relationship between $\hat{Y}$ and the residuals

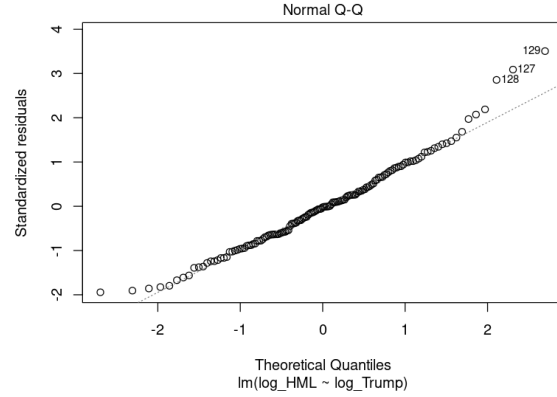


Figure 4.10: Q-Q plot of residuals

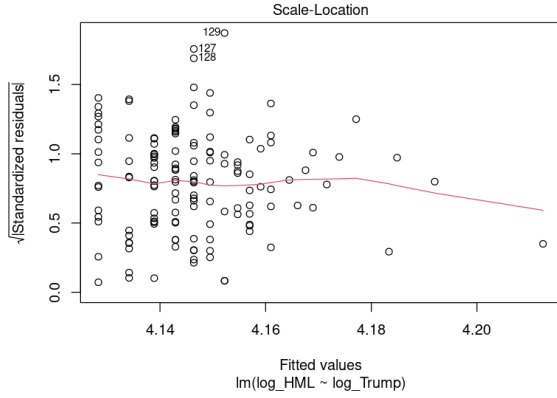


Figure 4.11: Relationship between $\hat{Y}$ and the standardized residuals

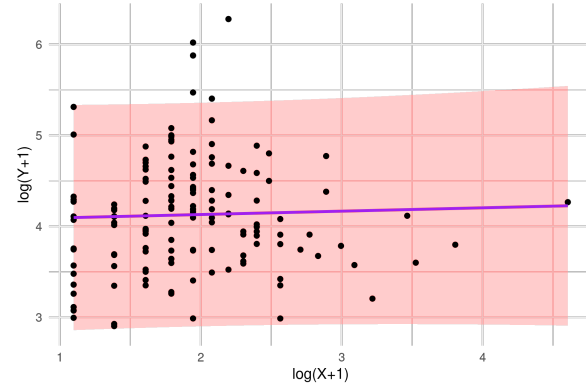


Figure 4.12: Linear regression line

Indeed, from the plots in Figures 4.9, 4.10, 4.11, 4.12, we see that the Gauss-Markov conditions are closer to being satisfied, so we can expect the estimators obtained by linear regression to be better. Linear regression gives $\log(Y + 1) = \hat{\beta} \log(X + 1) + \hat{\alpha}$, where $\hat{\beta} = 0.02607$ and $\hat{\alpha} = 4.0921$. To check the significance of this linear regression, we construct a 95% confidence interval for β , $[-0.1703286, 0.2224685]$, and observe that 0 lies within it. Moreover, the 90% confidence interval is $[-0.138419, 0.1995589]$, so even at the 0.10 level we cannot reject the hypothesis $H_0 : \beta = 0$ in favor of $H_1 : \beta \neq 0$.

Let us pay particular attention to this last result: we cannot obtain a statistically significant estimate of volume using only data on the frequency of searches for the word "Trump". In the next section, we will see how this changes when we add another regressor to the linear model.

4.2 Multiple linear regression

We now perform multiple regression with the independent variables “Trump” (X_1) and “Stocks” (X_2), and the dependent variable hml (Y).

Performing linear regression gives the relationship $Y = \hat{\alpha} + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$, with $\hat{\alpha} = -50.77$, $\hat{\beta}_1 = -1.08$, $\hat{\beta}_2 = 4.17$. We will test the model at the 0.05 significance level, and can now consider two sets of hypotheses:

$$H_0^1 : \beta_1 = 0,$$

$$H_1^1 : \beta_1 \neq 0$$

$$H_0^2 : \beta_2 = 0,$$

$$H_1^2 : \beta_2 \neq 0.$$

We construct 95% confidence intervals for β_1, β_2 in the same way as in the single-parameter case. For β_1 , we obtain $[-1.858206, -0.308047]$, and for β_2 , $[3.557971, 4.784710]$.

Neither the interval for β_1 nor the interval for β_2 contains zero. We conclude that we can reject both null hypotheses, meaning that in multiple regression, hml depends on both “Trump” and “Stocks”, although this was not the case in single-parameter regression. Slightly modified hypotheses are often used in multiple regression:

$$H_0' : \beta_1 = \beta_2 = 0,$$

$$H_1' : \beta_1 \neq 0 \vee \beta_2 \neq 0.$$

Our alternative hypotheses are strictly stronger than this one, in the sense that if we reject our null hypotheses H_0^1, H_0^2 , we necessarily also reject the null hypothesis H_0' . Thus, the preceding analysis also allows us to reject this null hypothesis. The interpretation is that performing regression was meaningful overall, since Y depends on the independent variables being examined.

In particular, looking only at $\hat{\beta}_1$, the coefficient of the variable “Trump”, we see that it is significant and negative. Thus, when the additional variable “Stocks” is taken into account, “Trump” becomes a significant predictor of hml, although it was not significant in the single-parameter case. Since the coefficient is negative, “Trump” and hml are inversely related. This is somewhat surprising, but we obtained a similar result (although not significant) in the previous section.

The 95% confidence interval for α is $[-72.344428, -29.192035]$.

The R^2 value is 0.5644. We have obtained a relatively good model for the data, for example compared with the models from single-parameter regression. Visualizing regression with two parameters is clearly more difficult, but we can still interpret it through Figure 4.13, which shows a relatively good relationship between the data and the model.

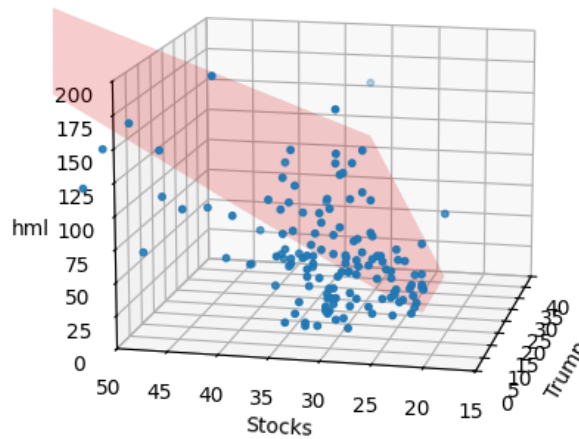


Figure 4.13: Multiple regression of the variables “Trump”, “Stocks”, and hml.

4.3 Kolmogorov-Smirnov test

We will use the Lilliefors version of the Kolmogorov-Smirnov test to check the distribution of hml. In Figure 3.4, the distribution appears log-normal; of all the variables examined, this histogram most strongly suggests a log-normal distribution. We therefore aim to test this claim at the 0.05 significance level. Let Y denote hml. We test the hypotheses:

$$H_0 : \log Y \sim N(\mu, \sigma)$$

$$H_1 : \log Y \text{ does not have a normal distribution.}$$

Applying the test to $\log Y$ gives $D = 0.05849$ and $p = 0.2701$. Since $p > 0.05$, we cannot

reject the null hypothesis that Y is log-normally distributed.

Let us apply the Lilliefors version of the Kolmogorov-Smirnov test to another example. Figure 3.11 suggests that change has a somewhat normal distribution. Denote it by X and test the hypothesis at the 0.05 significance level

$$H_0 : X \sim N(\mu, \sigma)$$

$$H_1 : X \text{ does not have a normal distribution.}$$

We obtain $D = 0.1148$ and $p = 8.739 \times 10^{-5}$. Since $p < 0.05$, we reject the null hypothesis and conclude that change is not normally distributed.

In general, examining other analyses based on much larger datasets suggests that the tails of the distribution of change are considerably heavier than those of a normal distribution. In our data, this is particularly apparent because of the outlier corresponding to the reciprocal tariffs.

5 Conclusion

In this seminar paper, we analyzed the relationship between searches for the terms "Trump" and "stocks" and indicators of the S&P 500 index. The descriptive statistics already suggested a linear relationship between trading volume and the frequency of searches for the word "stocks", and led us to make assumptions about the distribution of the difference between the highest and lowest daily prices (HML). This motivated us to perform linear regression and confirm its significance by calculating confidence intervals and using a t-test. Using the Lilliefors version of the Kolmogorov-Smirnov test, we did not reject the assumption that the distribution of HML is log-normal.

The relationship between the frequency of searches for the word "Trump" and HML was not clear from the descriptive statistics. Checking the Gauss-Markov conditions for linear regression of HML on the frequency of searches for the word "Trump" suggested a linear relationship between the logarithms of the two variables, but even then we could not establish statistical significance. However, adding the frequency of searches for the word "stocks" to the linear regression makes the effect of searches for the word "Trump" statistically significant. This shows that although information on the frequency of searches for the word "Trump" cannot explain HML, it is still useful provided that we have information on searches for the word "stocks".

Overall, we confirmed our expectation that performing statistical analysis on data concerning financial instruments is very challenging. Many factors are difficult to analyze meaningfully (news, changes in government, and so on), which is why all the confidence intervals are quite wide.

6 References

- Source of financial data: Yahoo Finance ^GSPC
- Source of search data: Google Trends
- 2025 stock market crash: Wikipedia
- Liberation Day: Wikipedia
- Q-Q plot: Wikipedia
- S&P500 Daily Returns Normal and Cauchy Fit